Mst Afsana Mimi

Amherst, MA, USA | mmimi@umass.edu | +1 207 437 8137 | github.com/mstafsanamimi

linkedin.com/in/mst-afsana-mimi-578234b1/ | mstafsanamimi.github.io

Summary

Statistics master's graduate with research experience developing and evaluating machine-learning models, engineering features, and preparing complex datasets for analysis. Proficient in Python, SQL, and R, with a record of translating statistical results into clear conclusions. Interested in applying rigorous modeling and careful data investigation to fraud, identity, and financial-risk problems.

Education

University of Massachusetts Amherst
Master's in Statistics (CGPA: 3.77) Sep 2023 – May 2026

University of Dhaka
Master's in Applied Statistics (CGPA: 3.75) Jan 2017 – Dec 2018

University of Dhaka
Bachelor's in Applied Statistics (CGPA: 3.61) Jan 2013 – Dec 2016

Technical Skills

Programming & Data Tools

- Python (Pandas, NumPy, Scikit-learn), SQL (joins, aggregations, window functions), R
- Git/GitHub; Tableau; AWS (S3, EC2) – familiarity

Machine Learning & Statistics

- Regression, Gaussian Processes, statistical modeling, hypothesis testing
- Feature selection, model development, experimentation, and performance evaluation

Data Workflows

- Data preprocessing, feature engineering, dataset construction, and reproducible analysis
- Data validation, consistency checks, and analytical data pipelines

Experience

Teaching Assistant (Data & Statistics)
UMass Amherst Sep 2023 – May 2026

- Supported student projects involving R/Python-based data analysis.
- Explained statistical methods and helped students communicate evidence-based conclusions clearly.

Research Assistant (NLP & Data Analysis)
University of Dhaka Jul 2021 – Jun 2022

- Prepared large-scale text datasets and built reproducible NLP workflows for ingestion, transformation, and analysis.
- Developed and evaluated machine-learning models in Python, including feature engineering and performance assessment.
- Applied cleaning, validation, and consistency checks to improve the reliability of analytical datasets.

Research Assistant (Socio-economic Data Analysis)
University of Dhaka Jul 2018 – Jun 2019

- Cleaned, transformed, and structured large-scale survey data for reproducible statistical analysis.
- Applied statistical models to investigate socio-economic patterns and identify decision-relevant findings.
- Maintained analysis code with Git and communicated results for policy-focused research.

Projects

Can Agentic AI Eliminate the Need for a Dedicated Relational Foundation Model?

- Developing an agentic machine-learning framework for relational databases to automate schema understanding, relational evidence construction, and predictive reasoning over structured data.
- Evaluating relational prediction workflows and developing a future benchmark for testing agentic systems on realistic database tasks.

Gaussian Process Modeling with Fairness Constraints

- Developing Gaussian Process models with constraints intended to reduce predictive disparities across demographic groups.
- Designing experiments to assess model accuracy, uncertainty, and fairness trade-offs.

High-Dimensional Feature Selection (MCAR)

- Developed a feature-selection method for high-dimensional censored data and tested it through simulations and biomedical data.
- Compared predictive-variable selection performance against established approaches; results were published in *Statistics in Medicine*.

Awards and Scholarship

- Conference Awards for Scientists (CASc) by International Society for Clinical Biostatistics (ISCB).
- National Science and Technology (NST) Fellowship Award by Education Board of Bangladesh.
- Dhaka University Merit Scholarship for academic results in B.S. Hon's examination.

Publications

- Mimi, A., & Khan, M. H. R. (2021). Variable selection for censored data using modified correlation adjusted correlation (mcar) scores. Statistics in Medicine, 40(23), 5046–5064.

Google Scholar: https://scholar.google.com/citations?user=LaVyjwYAAAAJ&hl=en&oi=sra